

Directional Collapse: One-Way Fine-Tuning Erases the Way Back

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Abstract

Many NLP tasks come in pairs whose training instances are the same data read in two directions: obfuscation and deobfuscation, translation each way, text-to-SQL and SQL-to-text, data-to-text and text-to-data. Fine-tuning on one direction destroys the ability the model already had in the other. We characterise this across `<<n-domains>>` paired domains and `<<n-families>>` model families, and show three things about it. The loss is disproportionate: the inverse direction falls to `<<abstract-reverse-after>>` while held-out general ability moves `<<abstract-general-cost>>`. It is cheap to prevent: replacing `<<abstract-knee>>` of the forward pairs with their own reversal restores `<<abstract-restored>>` of the lost capability, at matched instances, sequence tokens and optimizer steps. And what is lost is suppressed rather than erased — recoverable by prompting, by scaling the adapter, by a few reversed examples, and by a closed-form filter on the weight update that uses no data at all. Invertibility bounds the repair, and separates two things usually run together: an inverse that is undetermined cannot be restored at any dose, while one that is merely hard to find can be. A consequence follows for attribution: any method credited with restoring an inverse capability on paired data must be compared against a baseline supplying the same directional content and none of the method.

1 Introduction

Paired data carries a direction. Obfuscating a program rewrites it so that it behaves identically but is hard to read; a model can be trained to perform that rewrite, or to undo it. The two tasks use the same pairs of programs, and training the second only requires swapping which half of each pair is the input. Compilation and decompilation, minification and beautification, translation between two languages, and the two directions of semantic parsing all work this way.

This structure is what makes the failure we study both possible and avoidable. Fine-tuning a model on the forward direction of such a pair degrades its ability in the reverse direction — an ability the base model already had. Nikiema et al. [2025] name this *cognitive specialization* and document it for code obfuscation; Zhu et al. [2024] report the same shape in machine translation, where fine-tuning with English on the target side damages translation into other languages. We take the phenomenon as established in those settings and ask what is true of it in general.

The loss is disproportionate, which is why it is easy to miss. `<<intro-disproportion-sentence>>`

It is cheap to prevent. `<<intro-dose-sentence>>`

Invertibility bounds the repair, and it is two properties, not one. An inverse can be *undetermined* — the information needed is not present in the input, and no quantity of reverse data can supply it — or *determined but hard to find*, where the information is there and search is the obstacle. These predict different things about a cure, and §6 separates them with a synthetic ladder that varies determinability directly and a domain that holds determinability fixed while varying search difficulty.

What is lost is suppressed, not erased. ⟨⟨intro-mechanism-sentence⟩⟩

A general test follows, and it needs no new data. For any method that adds instances to paired training data, train the arm that supplies the same directional content and none of the method. This is more than advice because the ablation arm is also the remedy: running it pays for itself whichever way the comparison falls. §8 applies it to ⟨⟨n-attrib-methods⟩⟩ published objectives.

2 Directional collapse, defined

A *paired task* is a pair of datasets (X, Y) and (Y, X) over the same instances, where the reverse training set is obtained from the forward one by exchanging input and target. We call $X \rightarrow Y$ the *forward* direction and $Y \rightarrow X$ the *reverse*. *Directional collapse* is a fall in reverse-direction success after fine-tuning on forward-direction data alone, measured against the untouched model.

Every domain has a strict per-instance criterion with an automatic check (§3). Two failure modes are tracked separately and never counted as success:

Echo the output reproduces its own input. This is not hygiene. In JSON-to-YAML conversion, JSON is valid YAML, so a model that copies its input *parses* and *compares structurally equal* to the target; without the echo guard the task is not a task. Echo rates rose under forward-only training in earlier work, so a criterion that admits them measures the criterion rather than the capability.

Off-target the output is in the wrong language, format, or notation.

Thresholds for continuous criteria are fixed from the *untouched* model’s own score distribution before any fine-tuned model is scored, and are recorded with the date and quantile that produced them.

3 Setup

Domains. ⟨⟨setup-domains-sentence⟩⟩ [table domains not generated yet]

Models. ⟨⟨setup-models-sentence⟩⟩

Arms and budget. The forward-only baseline (SFT) trains on forward pairs. Each MIX k arm replaces $k\%$ of those pairs with their own reversal, partitioned by pair identity so that no pair is seen in both directions. Because the reversal *replaces* rather than adds, instance count, sequence tokens and optimizer steps are matched to SFT at every dose, and the only quantity varying along the ladder is the share of pairs seen backwards. Golovneva et al. [2024] distinguish data-matched from compute-matched reverse training; replacement is matched on both simultaneously, which is what licenses the word “free” in §5 — scoped, there and throughout, to training budget.

Two controls separate directional loss from its alternatives. REPLAY replaces the same share with generic instruction data, so REPLAY — SFT is ordinary forgetting and MIX — REPLAY is what direction specifically buys. REV trains on reverse pairs only and is the ceiling every reverse number is read against; where REV is near zero the direction is not learnable in that cell and no null there is interpretable. [table arms not generated yet]

4 Collapse is general and disproportionate

⟨⟨general-headline-sentence⟩⟩ [table main not generated yet]

It is not ordinary forgetting. ⟨⟨general-replay-contrast⟩⟩

It is not a loss of instruction following. ⟨⟨general-ifeval-contrast⟩⟩

Moderators. ⟨⟨general-moderators⟩⟩

5 A small reversed dose restores it at no cost

⟨⟨dose-headline-sentence⟩⟩ [table dose not generated yet]

Where the knee is. We define the knee as the smallest dose whose reverse success reaches 90% of MIX50’s in that cell. ⟨⟨dose-knee-sentence⟩⟩

What it costs. ⟨⟨dose-cost-sentence⟩⟩ Fine-tuning costs general ability at any direction mix; the dose is free relative to the forward-only baseline, not relative to the untouched model.

6 Invertibility bounds what can be restored

⟨⟨invertibility-headline-sentence⟩⟩ [table ladder not generated yet]

Two kinds of hardness. ⟨⟨invertibility-two-kinds⟩⟩

7 Suppressed, not erased

Four instruments, reported as separate verdicts rather than averaged: they measure different things, and a disagreement between them is a finding rather than noise.

Prompting recovers part of what tuning removed. ⟨⟨mech-elicitation⟩⟩

The collapse is a cheap direction in weight space. ⟨⟨mech-alpha⟩⟩

Relearning outruns a capability never held. ⟨⟨mech-relearn⟩⟩ This is also where the contrast with the reversal curse [Berglund et al., 2024] becomes experimental rather than rhetorical: that literature describes capabilities never acquired, and our never-had control is that case, constructed.

A closed-form filter on the update restores it with no data. ⟨⟨mech-spectral⟩⟩

8 Consequences for attribution

⟨⟨attrib-headline-sentence⟩⟩

We do not claim that every auxiliary-objective result on paired data is a directional artifact. We measured ⟨⟨n-**attrib-methods**⟩⟩; the confound is real, cheap to test for, and currently untested, and anything stronger than that is unsupported.

9 Related work

⟨⟨related-work-section⟩⟩

10 Limitations

⟨⟨limitations-section⟩⟩

11 Conclusion

For any method that adds instances to paired training data, train the arm that supplies the same directional content and none of the method.

References

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- Dawei Zhu, Pinzhen Chen, Miaoran Zhang, Barry Haddow, Xiaoyu Shen, and Dietrich Klakow. Fine-tuning large language models to translate: Will a touch of noisy data in misaligned languages suffice? In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024. URL <https://aclanthology.org/2024.emnlp-main.24/>. Single-direction fine-tuning generalizes, EXCEPT that English on the target side causes task misinterpretation damaging translation into non-English.